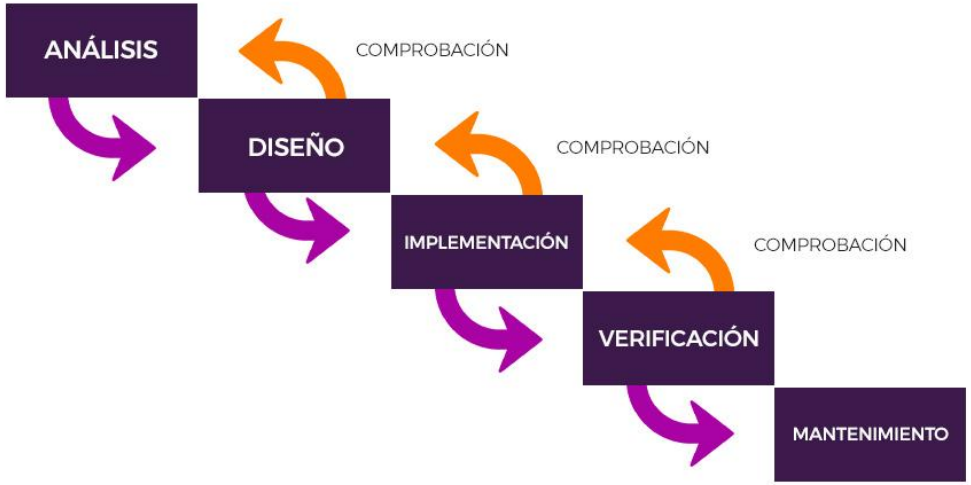
SEKUENTZIA DIAGRAMAK

Ander Madinabeitia
amadinabeitia@goierrieskola.org

GARAPEN PROZESUAK

- Softwarearen garapen prozesuan ondorengo faseak bereizten dira:
    - Analisia
    - **Diseinua**
    - Programazioa
    - Frogak
    - Martxan jartzea
    - Mantentze lanak

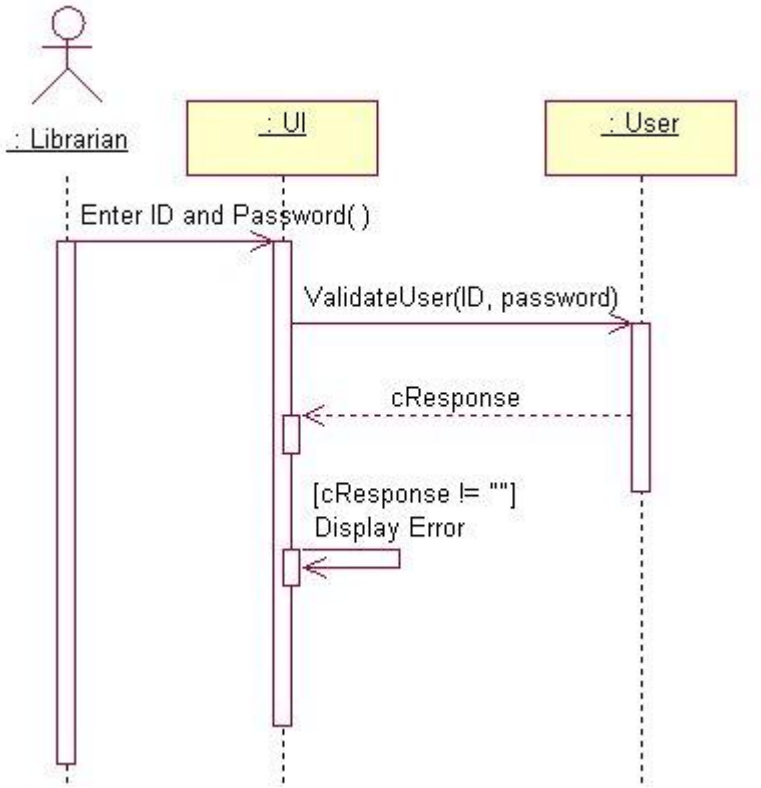


UML-DIAGRAMAK

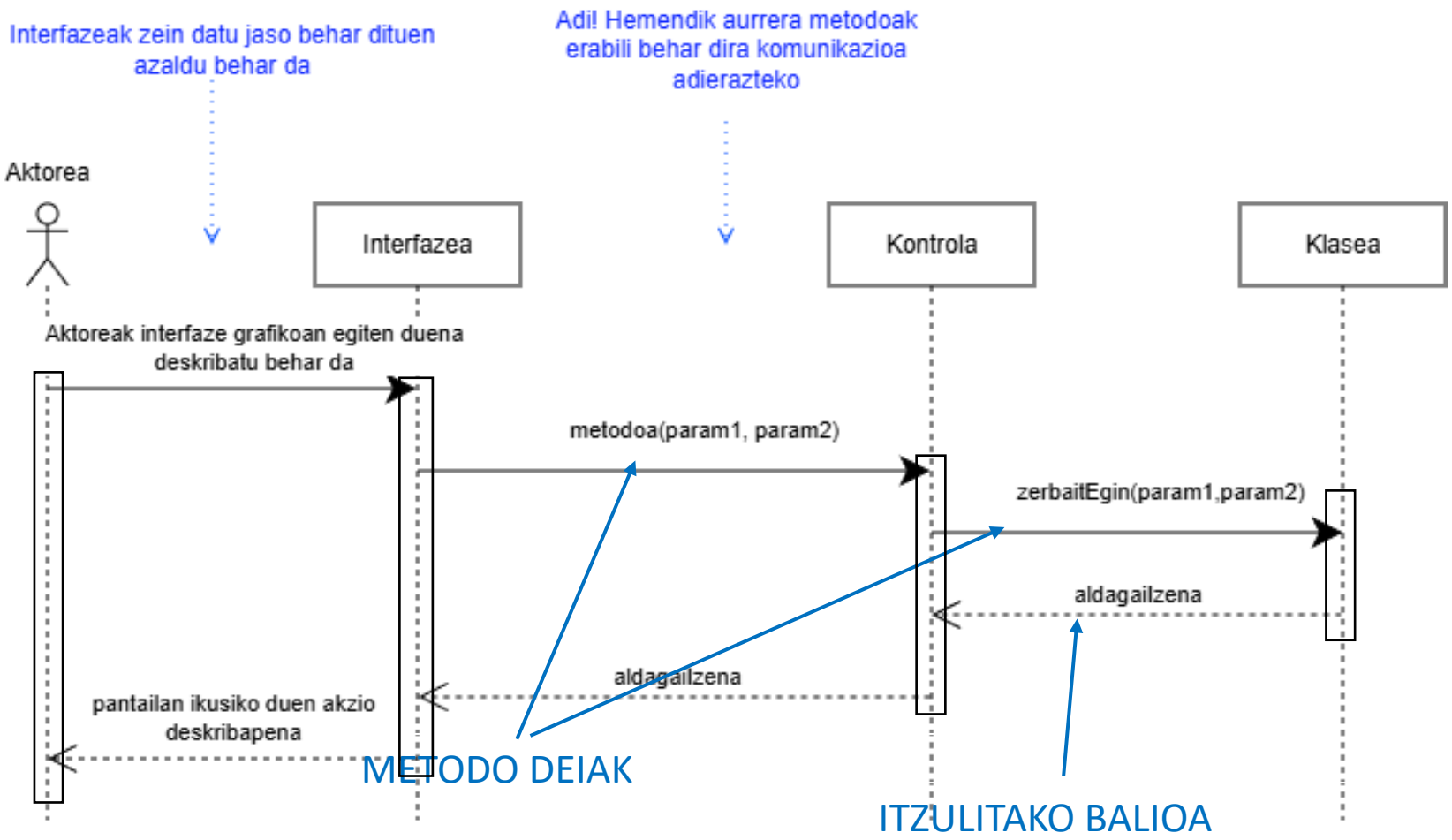
- Helburua gai konplexuak argi eta garbi aurkeztea da.
- UMLk notazio estandarizatu bat ezartzen du eta forma bisualak erabiltzen ditu osagai edo portaera espezifiko bat irudikatzeko.

SEKUENTZIA DIAGRAMAK

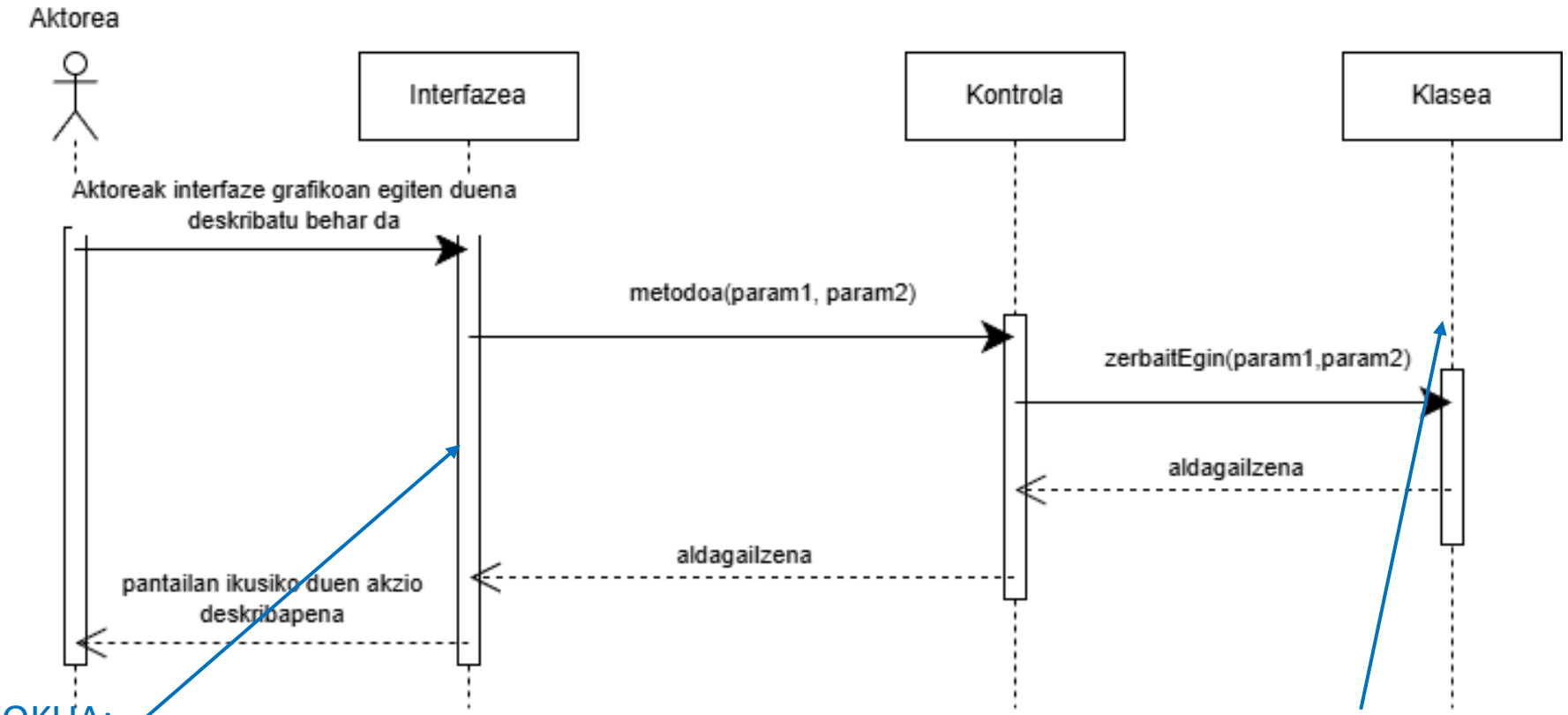
- Sekuentzia-diagramek, erabilpen-kasu baten dauden **objektuen** artean **trukatutako** mezuen sekuentzia **kronologikoa** erakusten dute.
- Sisteman eskaerak nola egiten diren eta mezuak nola itzultzen diren adierazten dute.
- Grafikoki azaltzen dituzte eragiketak.



SEKUENTZIA DIAGRAMAK (I)



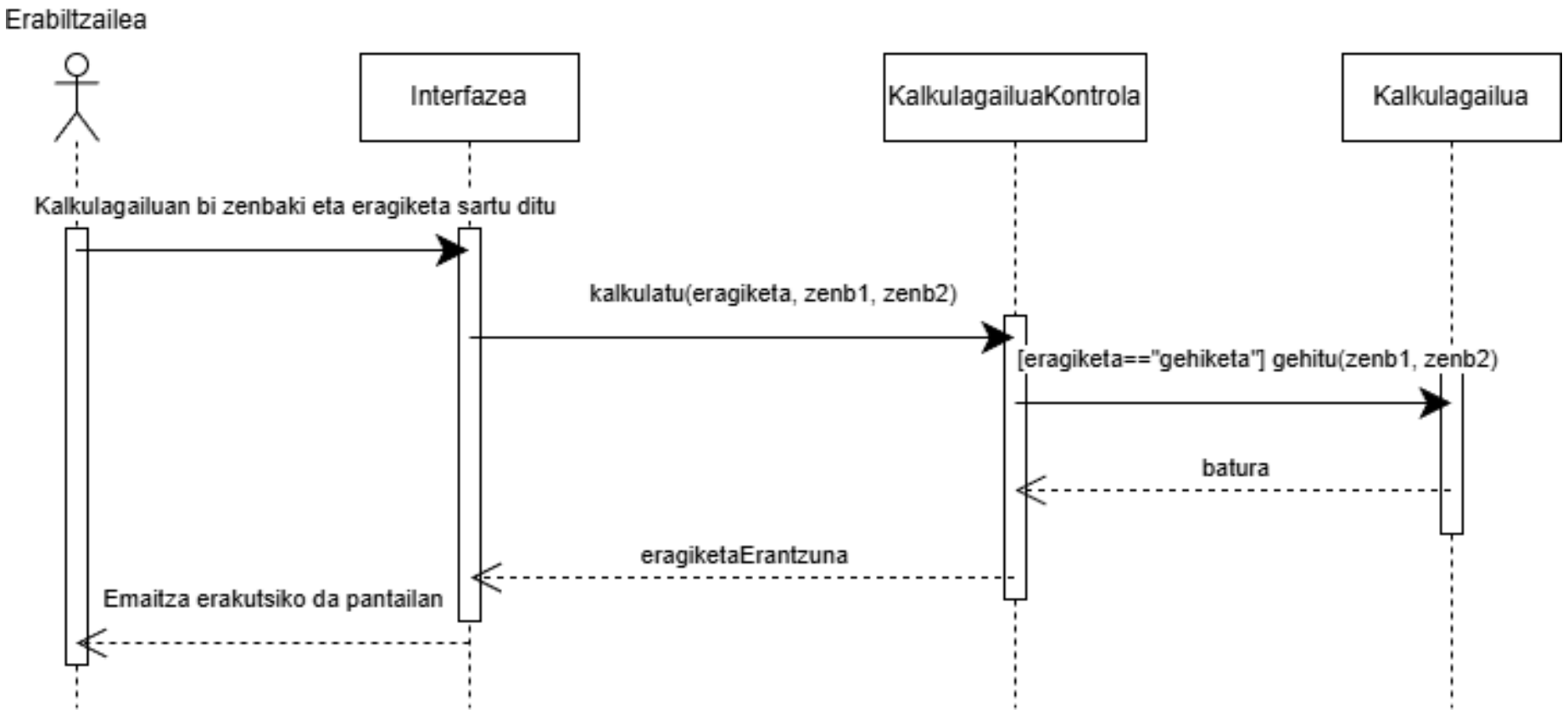
SEKUENTZIA DIAGRAMAK (II)



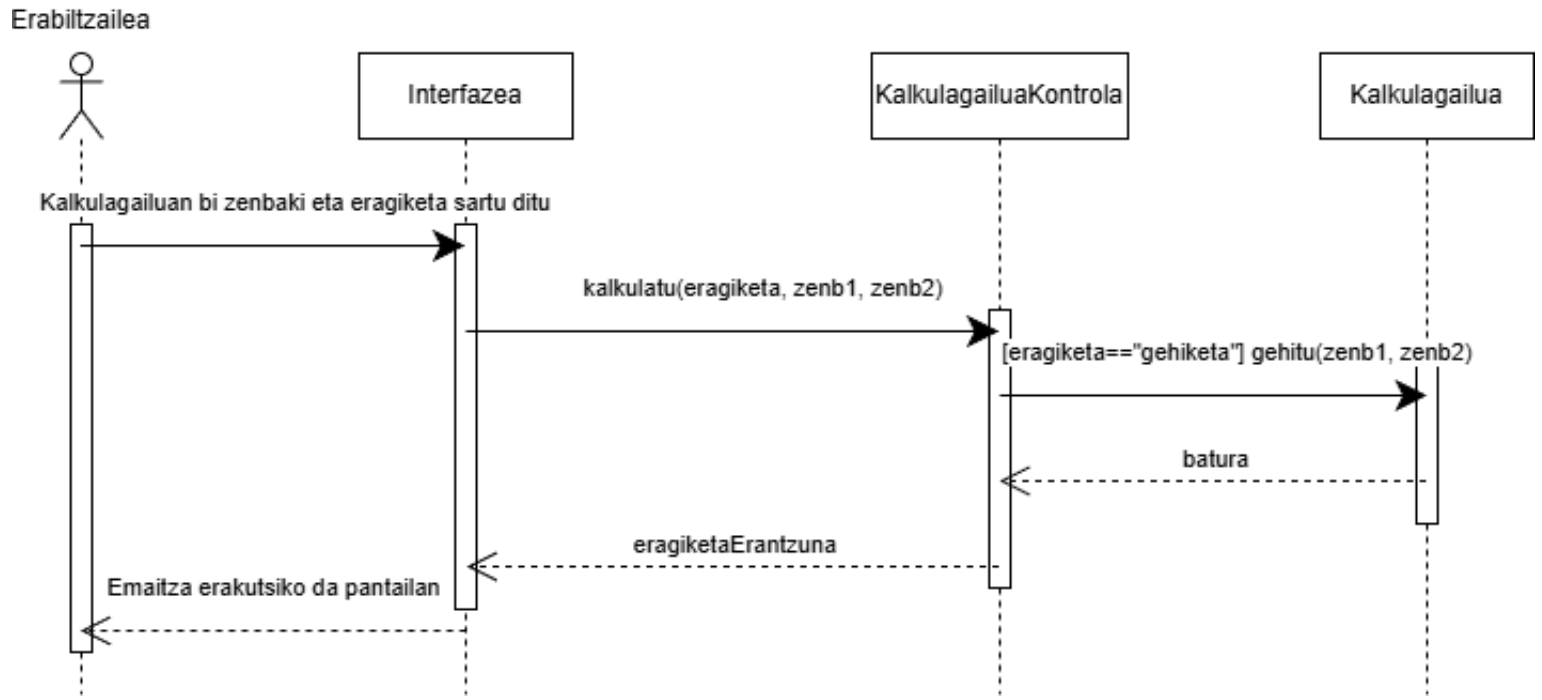
KONTROL FOKUA:
Objektuak ondorengo akzioa egiteko behar duen denbora

BIZITZAREN LERROA:
Objektua bizirik dagoen denbora

SEKUENTZIA DIAGRAMAK - ADIBIDEA

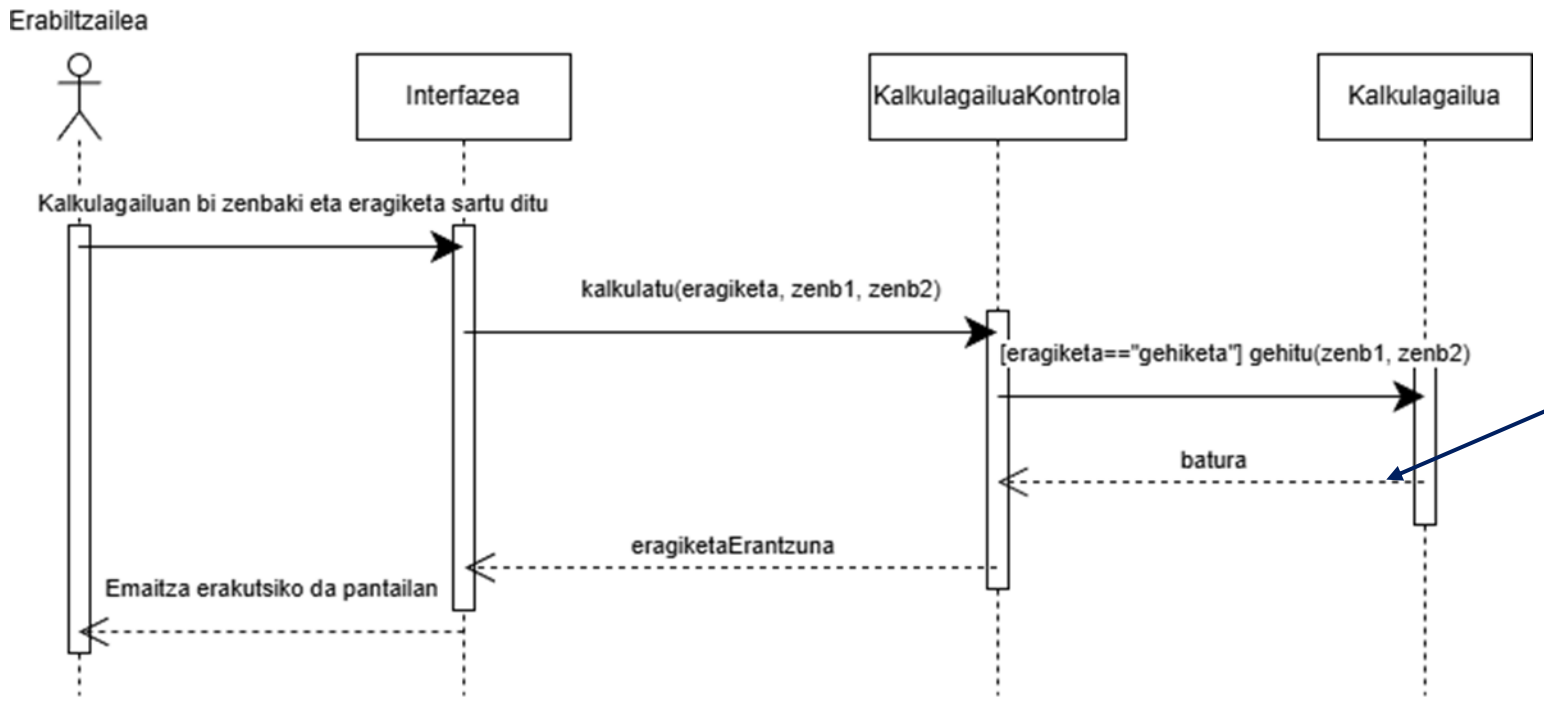


SEKUENTZIA DIAGRAMAK (III)



Kalkulagailua klasean gehitu(zenb1, zenb2) funtzioa existitu behar da.

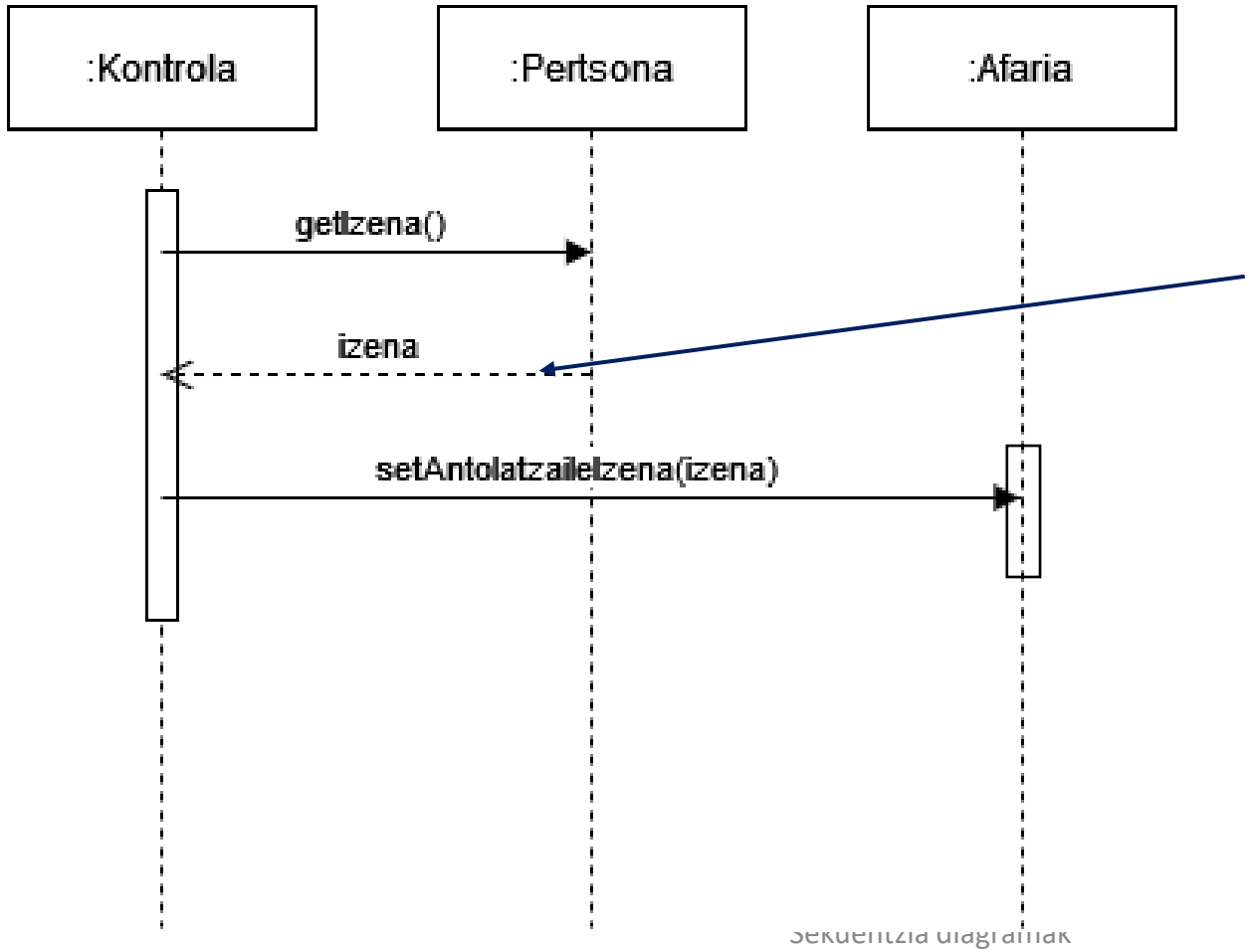
SEKUENTZIA DIAGRAMAK (IV)



Metodo-dei batek **balio bat itzuli dezake**.

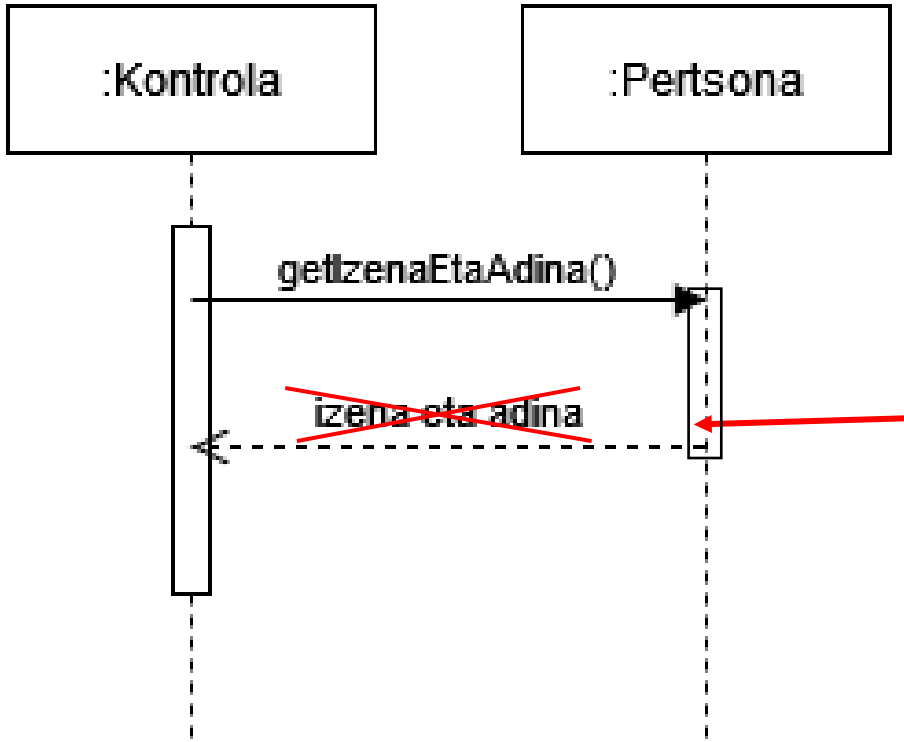
Soilik informazio interesgarria itzultzen badu adieraziko da. Bestela posible da ez adieraztea.

SEKUENTZIA DIAGRAMAK (V)



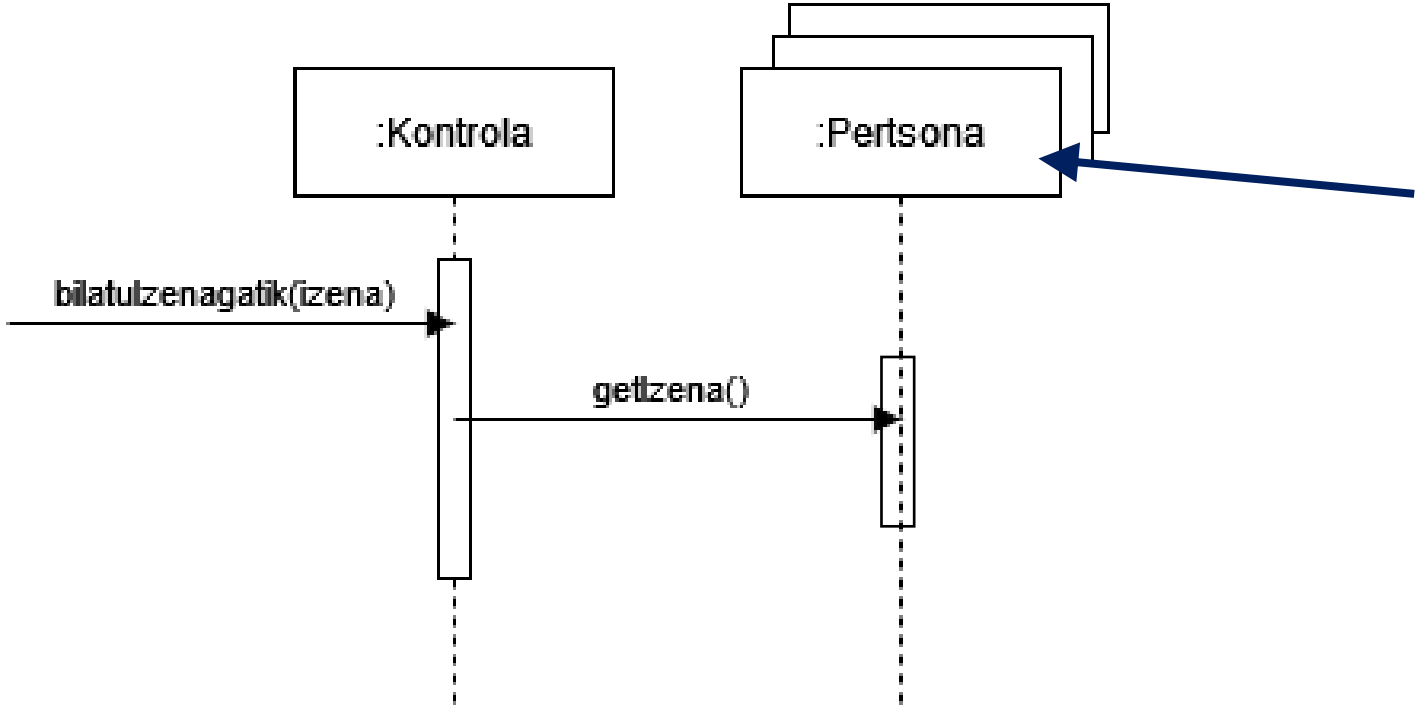
Kasu honetan BAI beharrezkoa da. Horrela ikusi dezakegu Pertsona-tik zein datu hartzen den eta non erabiltzen den datu hori.

SEKUENTZIA DIAGRAMAK (VI)



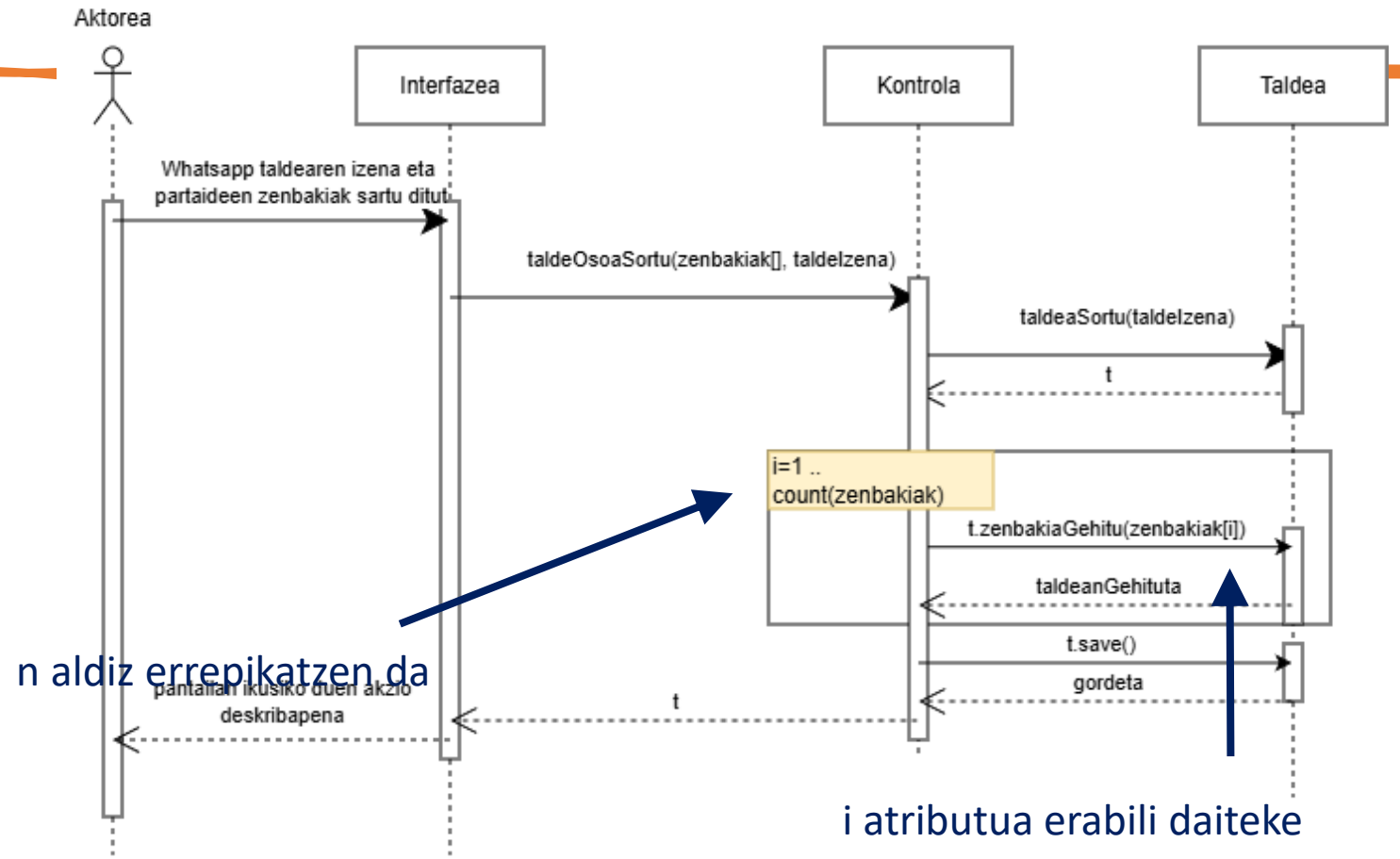
Metodo dei batek EZIN DU balio bat baino gehiago itzuli

SEKUENTZIA DIAGRAMAK (VII)



Pertsona klasearen objektu anitzak

SEKUENTZIA DIAGRAMAK (VIII)



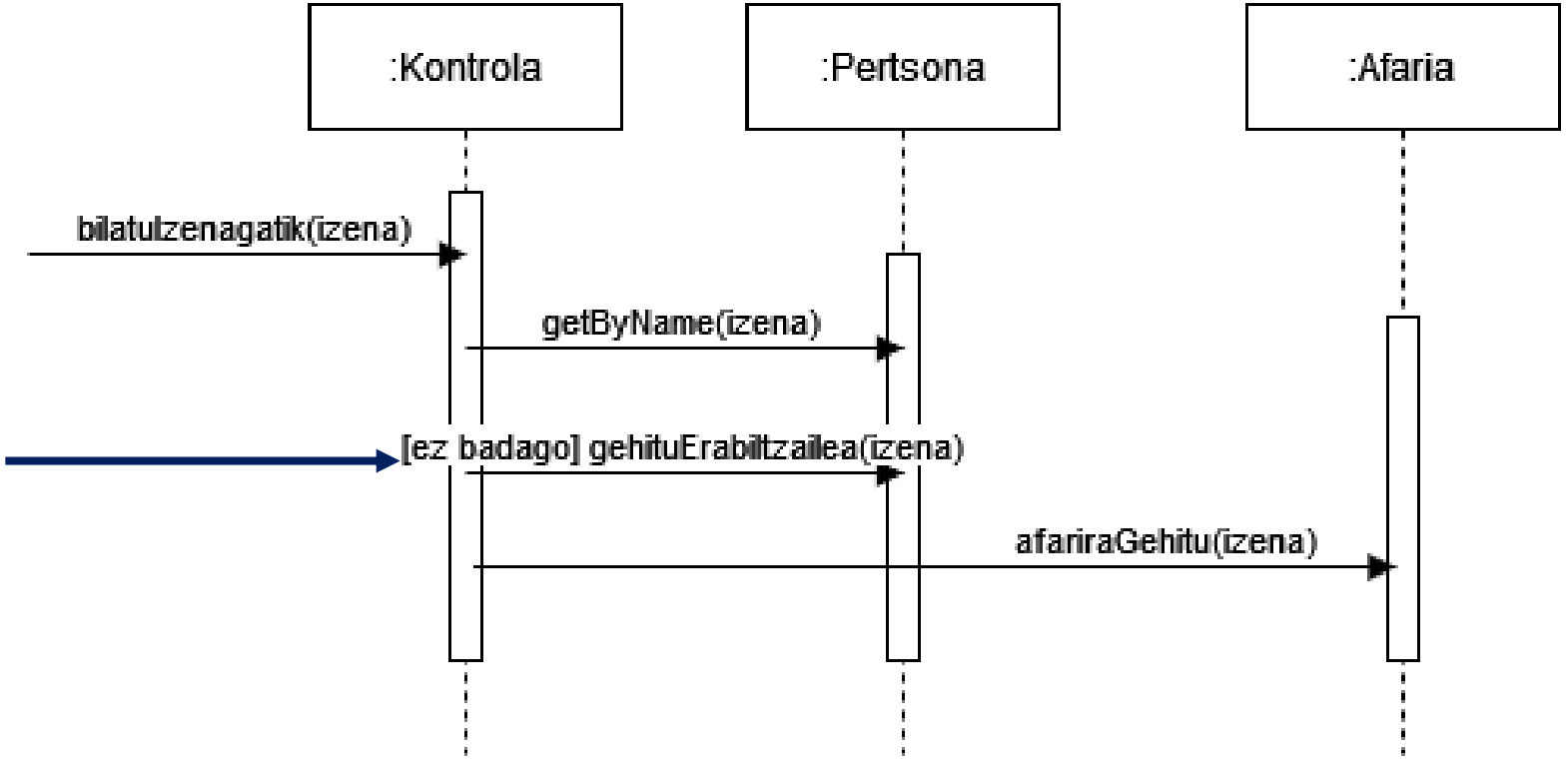
n aldiz errepikatzen da

i atributua erabili daiteke

ERREPIKAPENA:

Honela errepikatu nahi diren mezu-pasatze guztiak lauki batean sartzen dira. Azaldu daiteke: zenbat aldiz errepikatuko den edo noiz arte errepikatuko den adierazten den baldintza

SEKUENTZIA DIAGRAMAK (IX)



Baldintzak ere sartu daitezke "[]" artean

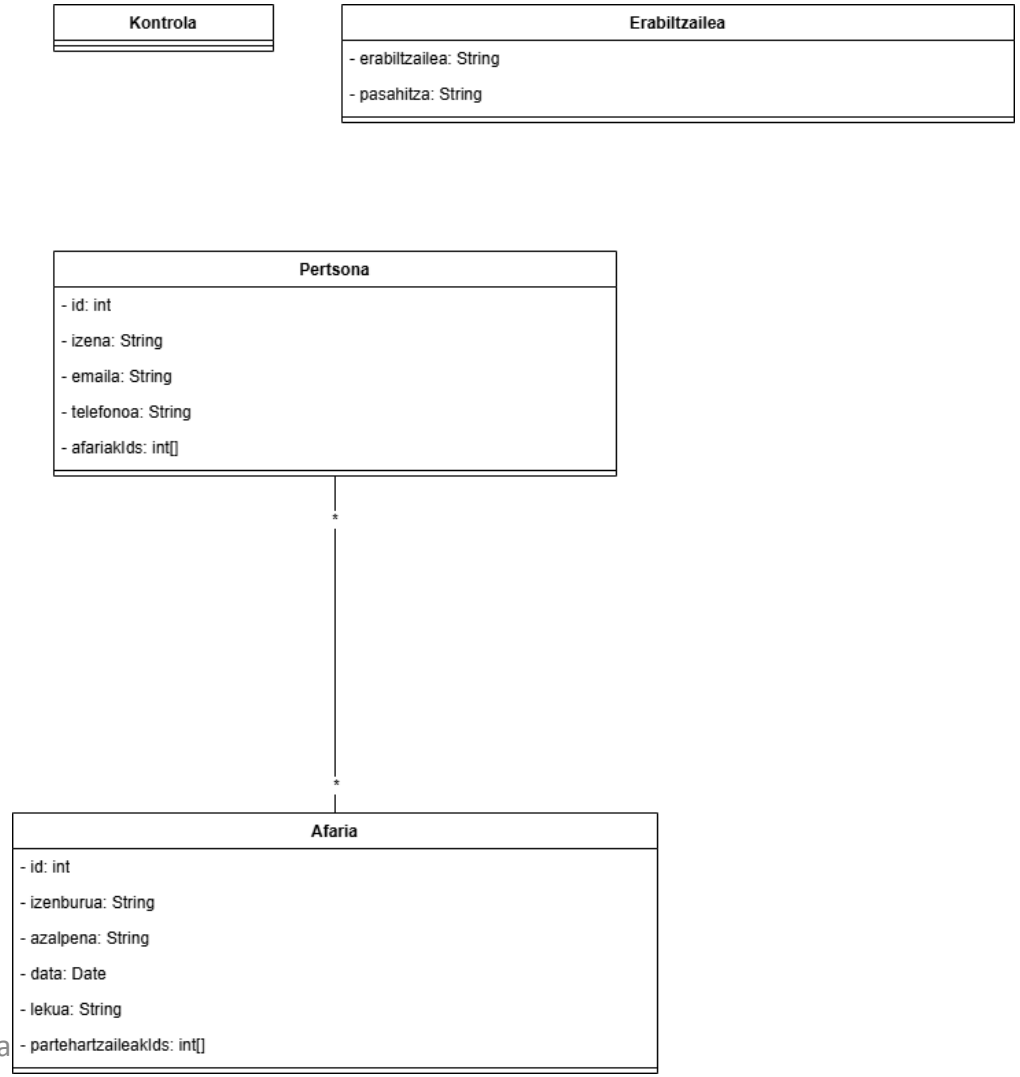
AURKEZPENA, NEGOZIO LOGIKA ETA DATU ATZIPENA

- Aurkezpena: Interfaze grafikoarekin erlazionatutako eragiketak.
- Negozio logika (Kontrola): EKko logikarekin erlazionatutako eragiketak.
- Datu atzipena: Objektuak sortu, eguneratu, ezabatu eta datuak berreskuratu.

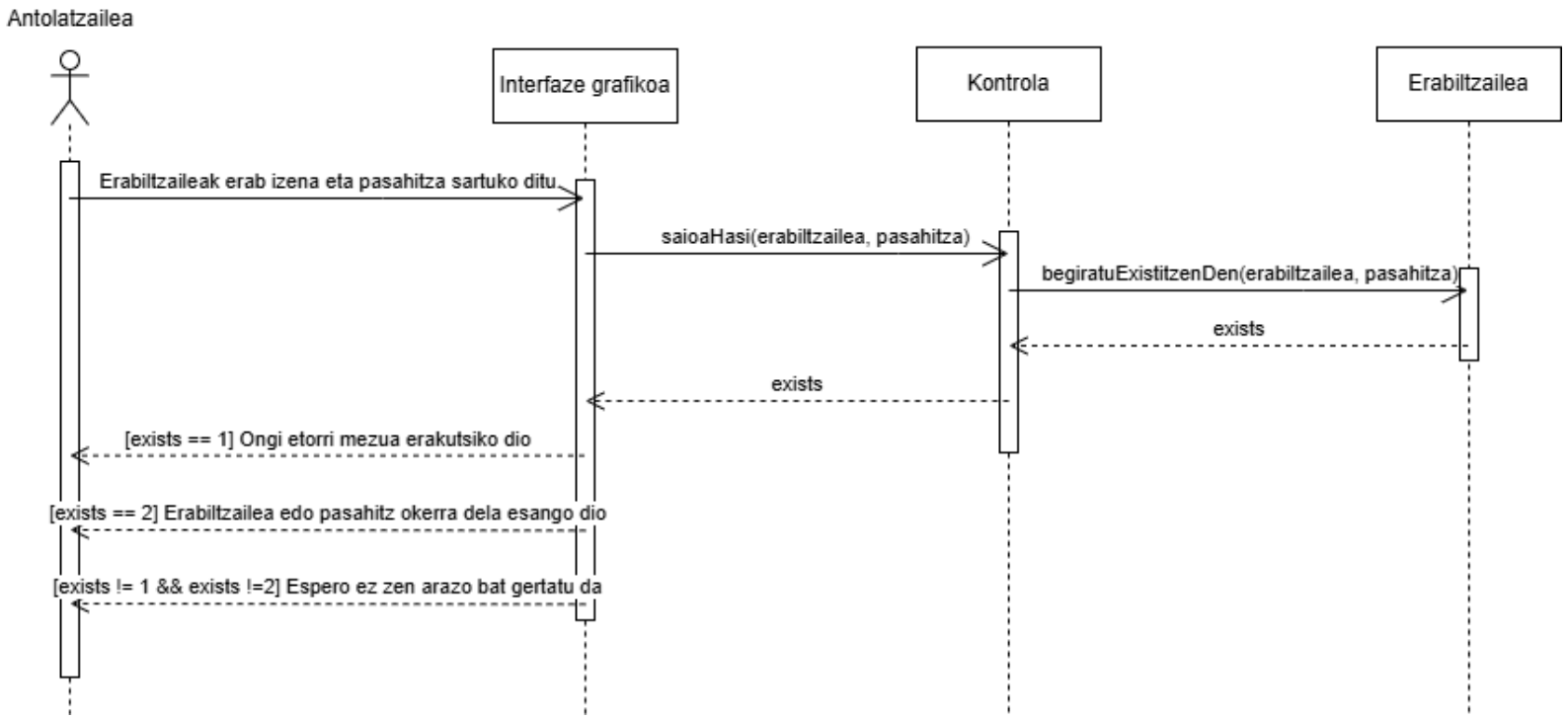
ADIBIDEA

Ondorengo erabilpen kasuak dauzkan sekuentzia diagramak sortu: Login egin, Afaria sortu, afaria ezabatu, afarian izena eman

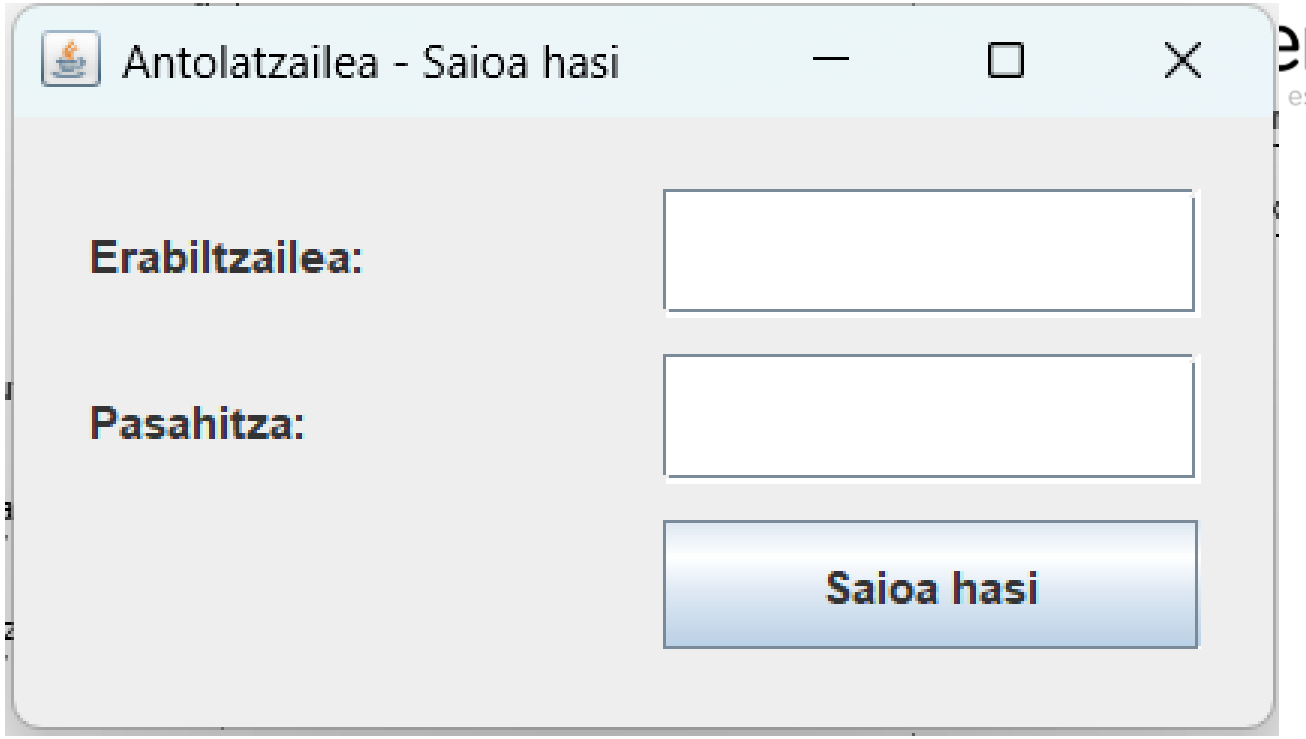
Klase diagrama ondorengoa da:



ADIBIDEA: LOGIN EGIN



ADIBIDEA: LOGIN EGIN



LoginFrame.java

```
9   public class LoginFrame extends JFrame {  1 usage
14      public LoginFrame() {  1 usage
31
32          JButton btnLogin = new JButton( text: "Saioa hasi");
33          btnLogin.addActionListener(new ActionListener() {
34              public void actionPerformed(ActionEvent e) { login(); }
37          });
```

ADIBIDE EGIN

LoginFrame.java

```java
public class LoginFrame extends JFrame {  1 usage

    private void login() {  1 usage
        String erabiltzailea = txtErabiltzailea.getText();
        String pasahitza = new String(txtPasahitza.getPassword());

        Kontrol controller = new Kontrol();

        switch (controller.saioaHasi(erabiltzailea, pasahitza)){
            case 1:
                JOptionPane.showMessageDialog( parentComponent: this,  message: "Ongi etorri, " + erabiltzailea + "!");
                // Ireki admin panel
                new AdminPanelFrame().setVisible(true);
                dispose();
                break;
            case 2:
                JOptionPane.showMessageDialog( parentComponent: this,  message: "Erabiltzailea edo pasahitza okerra",  title: "Errorea", JOptionPane.ERRO
                break;
            default:
                JOptionPane.showMessageDialog( parentComponent: this,  message: "Espero ez zen errore bat gertatu da, saiatu beranduago",  title: "Errore
                break;
        }


    }
```

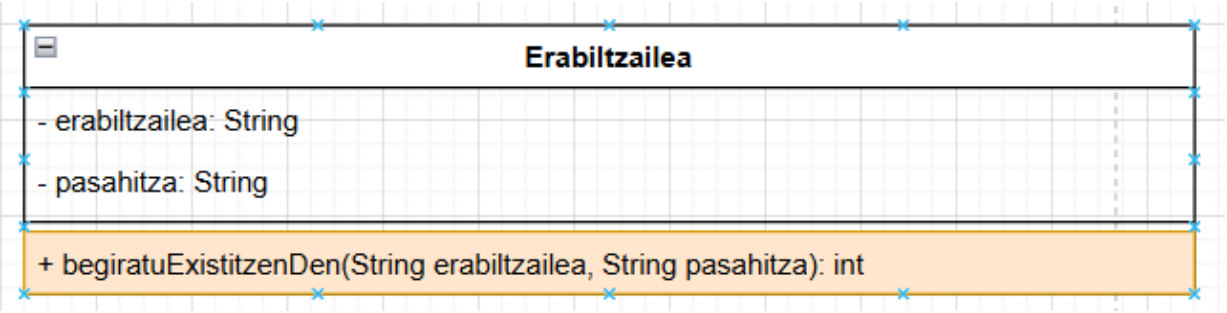
Goierri eskola

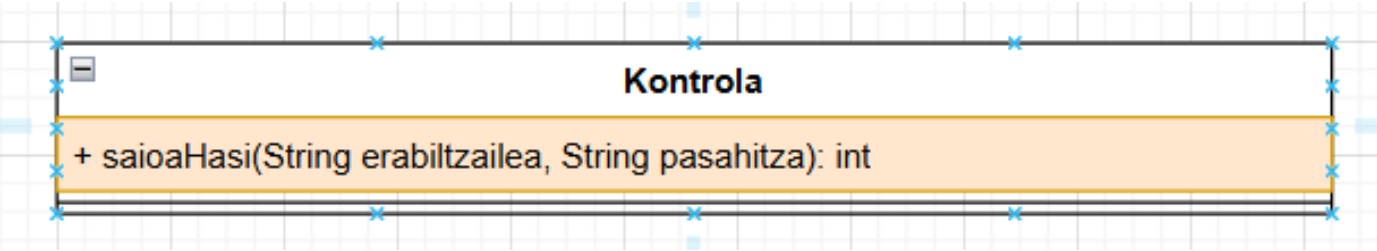
```java
public class Kontrol {  12 usages
    public int saioaHasi(String erabiltzailea, String pasahitza) {  1 usage
        return Erabiltzailea.begiratuExistitzenDen(erabiltzailea, pasahitza);
    }
```

ADIBIDE EGIN

```java
public class Erabiltzailea {  2 usages

    private String erabiltzailea;  no usages
    private String pasahitza;  no usages

    public static int begiratuExistitzenDen(String erabiltzailea, String pasahitza) {  1 usage
        try (Connection conn = DbKonexioa.konektatu();
             PreparedStatement stmt = conn.prepareStatement( sql: "SELECT pasahitza FROM erabiltzaileak WHERE erabiltzailea = ? AND pasahitza = ?")
            stmt.setString( parameterIndex: 1, erabiltzailea);
            stmt.setString( parameterIndex: 2, pasahitza);
            ResultSet rs = stmt.executeQuery();

            return rs.next() ? 1 : 2; //Existitzen bada 1, bestela 2

        } catch (SQLException ex) {
            ex.printStackTrace();
            return -1;
        }
    }
}
```

ADIBIDEA: LOGIN EGIN





ADIBIDEA: AFARIA SORTU

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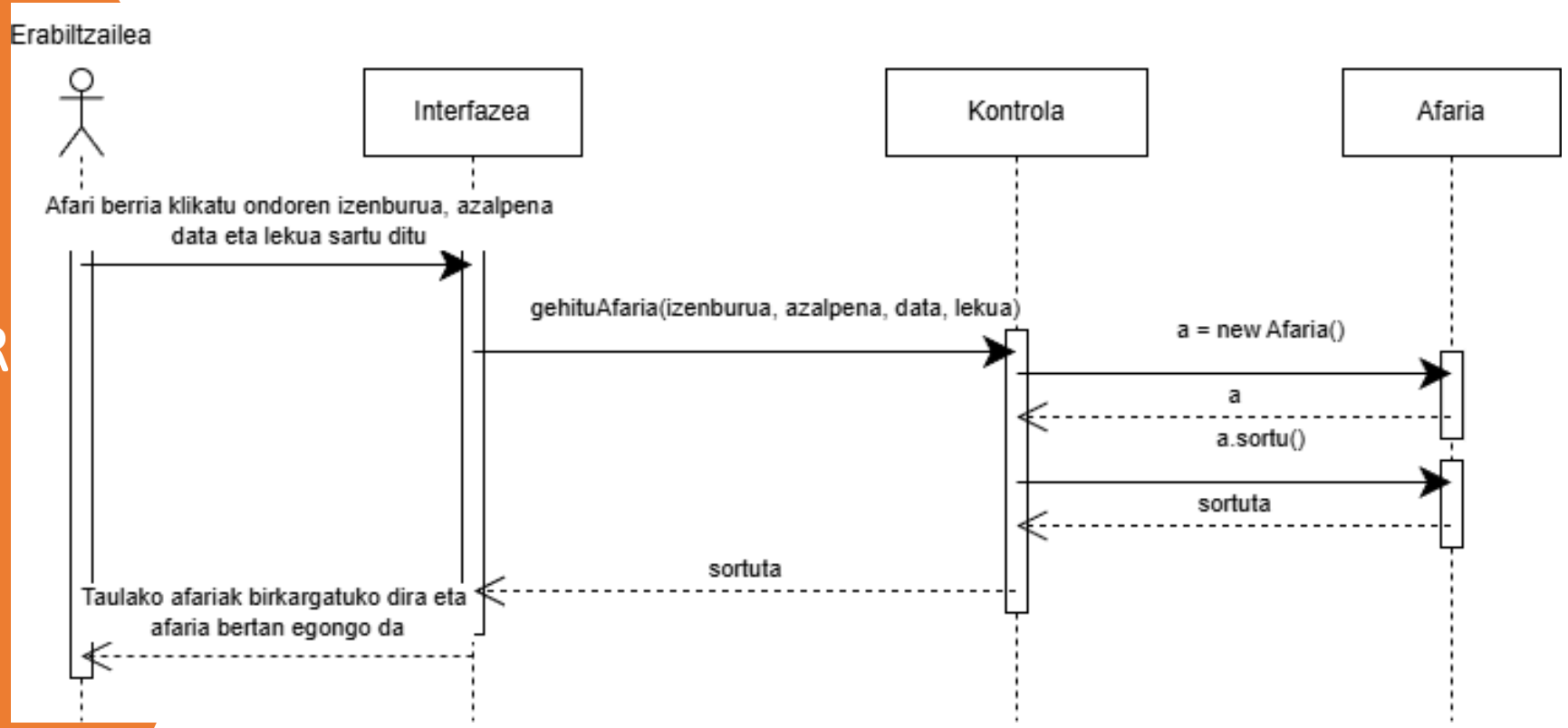

ADIBIDEA: AFARI
SORTU

```java
JButton btnSave = new JButton( text: "Gorde");
btnSave.addActionListener(this::saveAfaria);
add(btnSave, BorderLayout.SOUTH);
```

```java
private void saveAfaria(ActionEvent e) {  1 usage
    try {
        String izenburua = txtIzenburua.getText();
        String azalpena = txtAzalpena.getText();
        Date data = (Date) txtData.getValue();
        String lekua = txtLekua.getText();

        if (afariaId == null) {
            controller.gehituAfaria(izenburua, azalpena, data, lekua);
        } else {
            controller.eguneratuAfaria(izenburua, azalpena, data, lekua, afariaId);
        }

        parent.loadAfariak();
        dispose();

    } catch (Exception ex) {
        ex.printStackTrace();
        JOptionPane.showMessageDialog( parentComponent: this,  message: "Errorea gordetzean",  title: "Erro
    }
}
```

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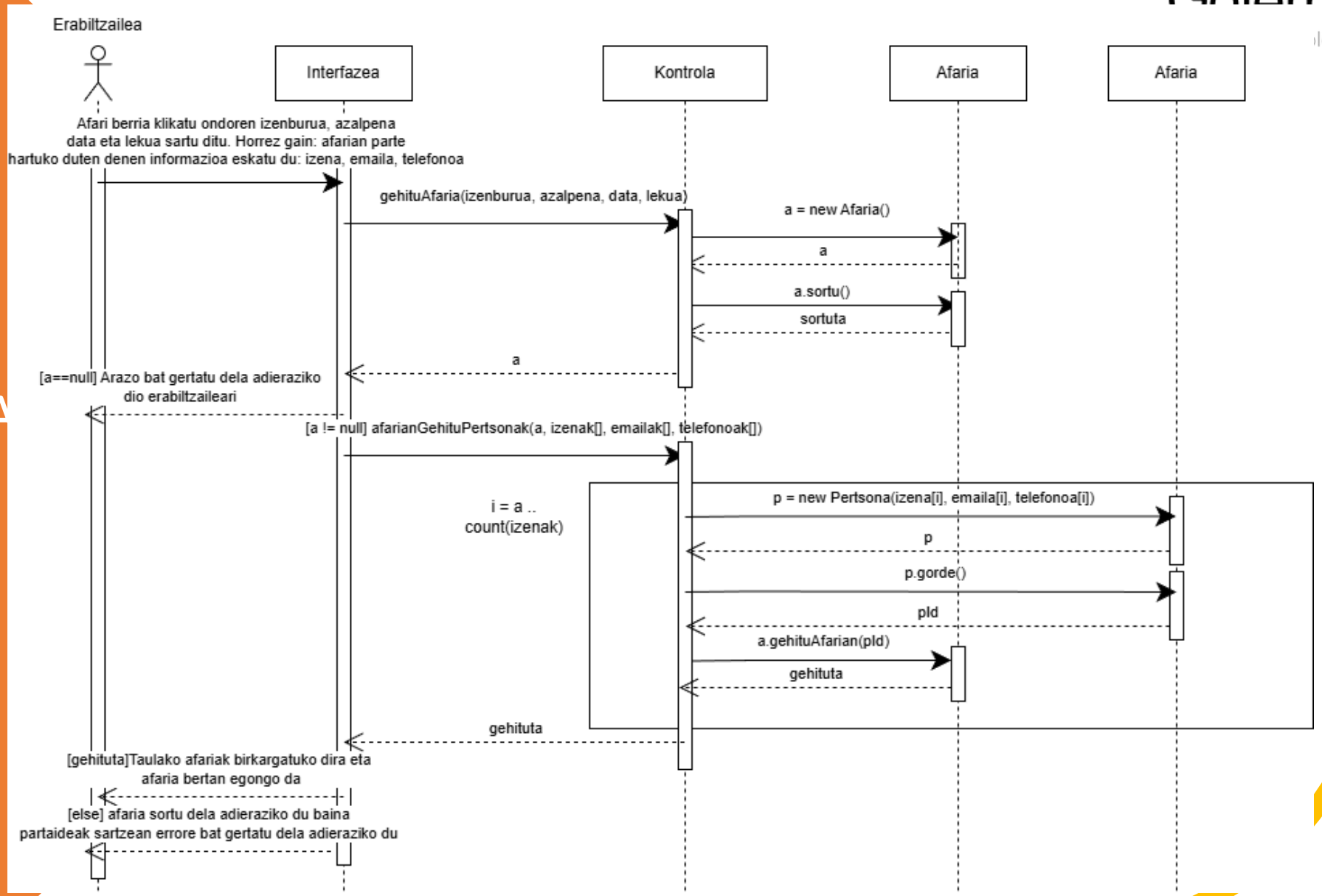
SOF

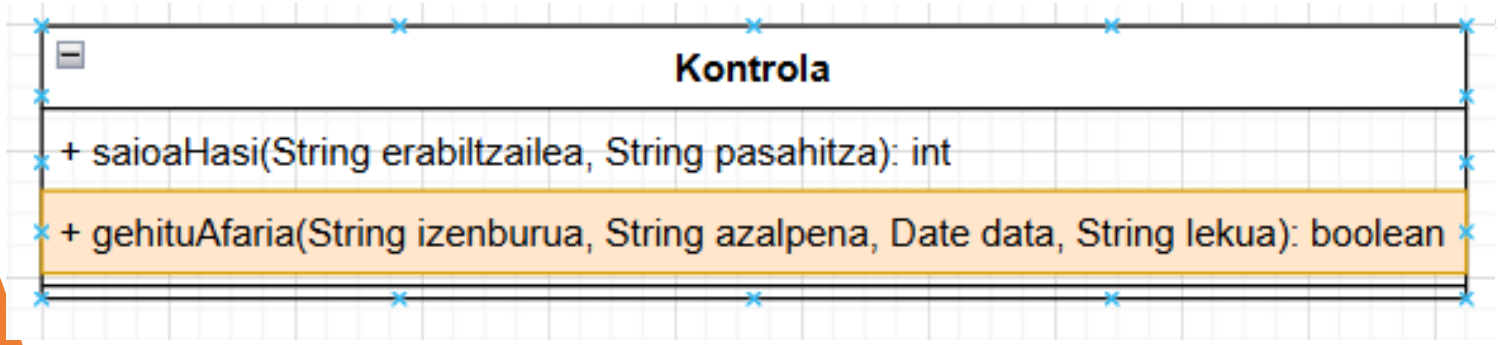
```
Change signature
public boolean gehituAfaria(String izenburua, String azalpena, Date data, String lekua) throws SQLException {
    Afaria a = new Afaria();
    a.setIzenburua(izenburua);
    a.setAzalpena(azalpena);
    a.setData(data);
    a.setLekua(lekua);

    return a.sortu();
}
```

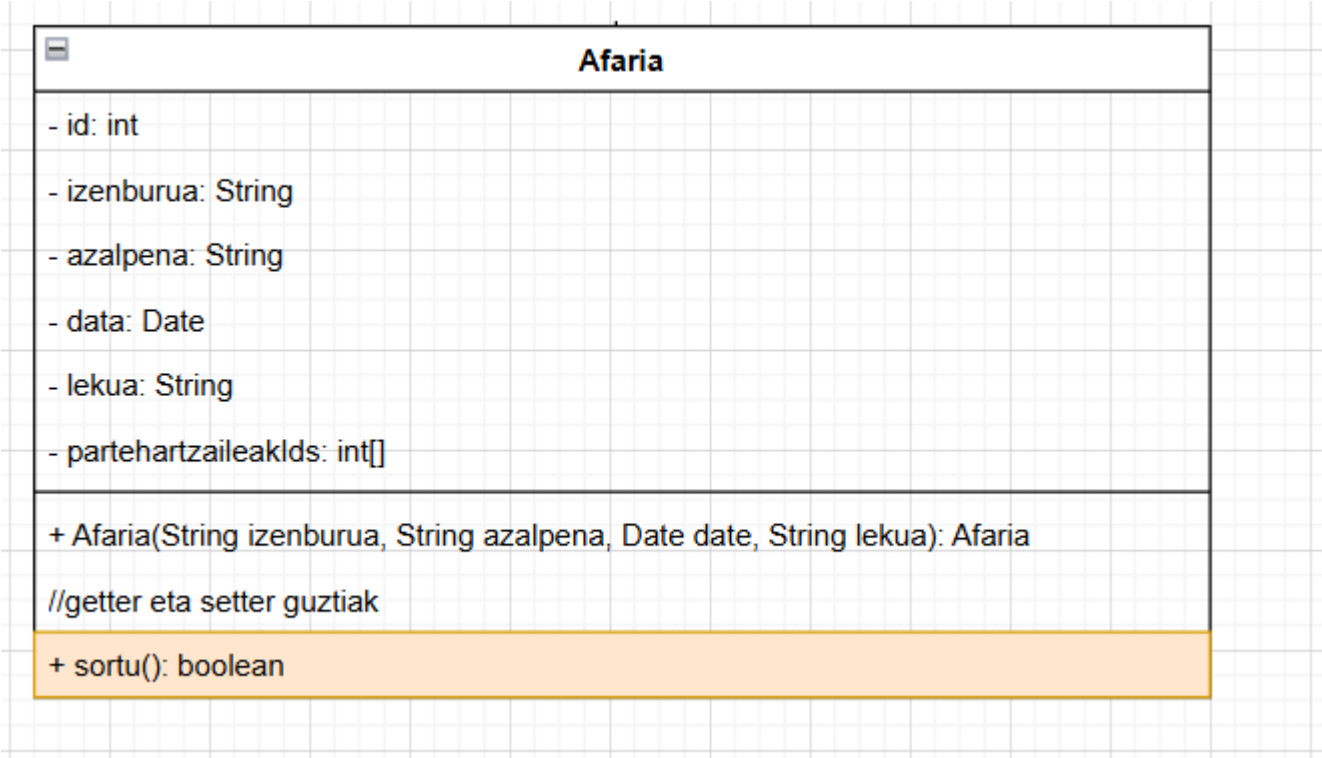
```
public boolean sortu() throws SQLException {  1 usage
    String query = "INSERT INTO afariak (izenburua, azalpena, data, lekua) VALUES (?, ?, ?, ?)";
    try (Connection conn = DbKonexioa.konektatu(); PreparedStatement stmt = conn.prepareStatement(query)) {
        stmt.setString( parameterIndex: 1, this.getIzenburua());
        stmt.setString( parameterIndex: 2, this.getAzalpena());
        stmt.setDate( parameterIndex: 3, new java.sql.Date(this.getData().getTime()));
        stmt.setString( parameterIndex: 4, this.getLekua());
        stmt.executeUpdate();
        return true;
    } catch (SQLException e) {
        e.printStackTrace();
        return false;
    }
}
```

ADIBIDEA: AFA
SORTU



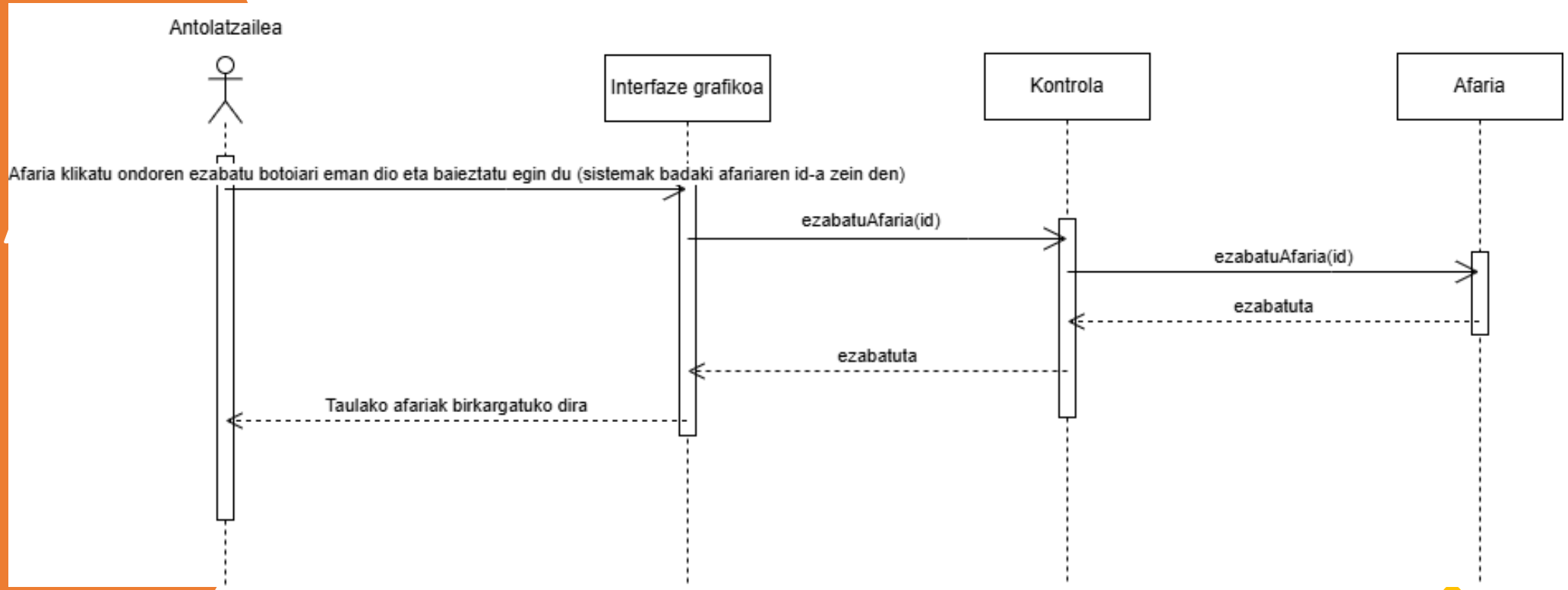
ADIBIDEA: AFARIA SORTU



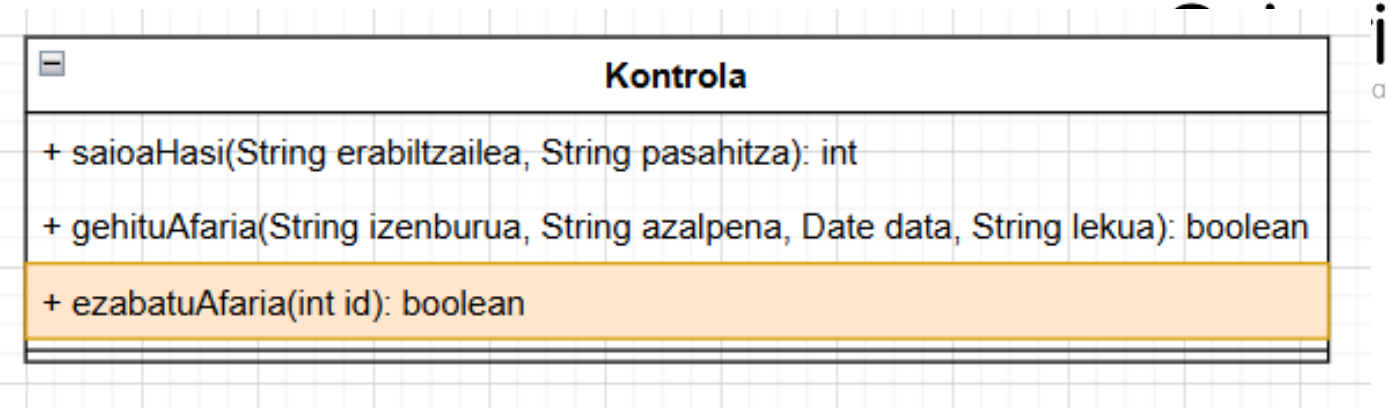


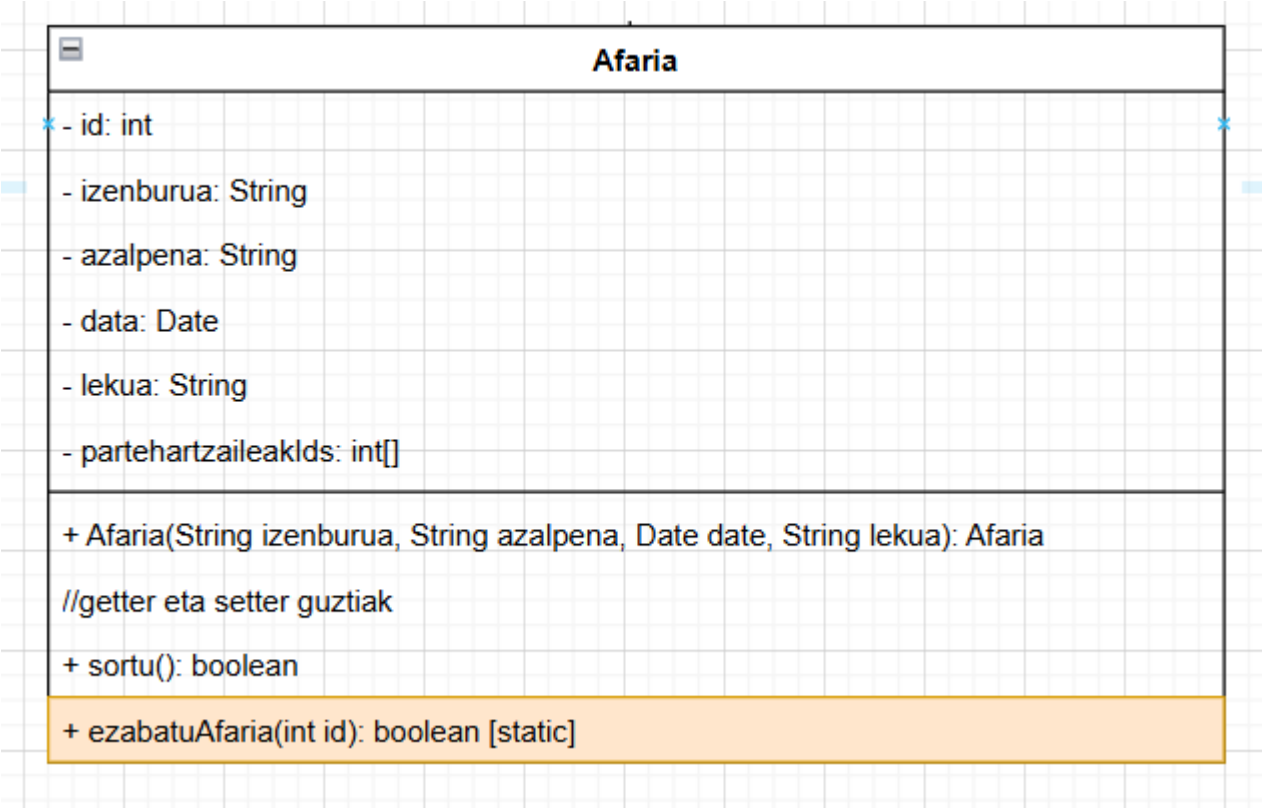
ADIBIDEA: AFARIA EZABATU

ADIBIDEA: A
EZABATU



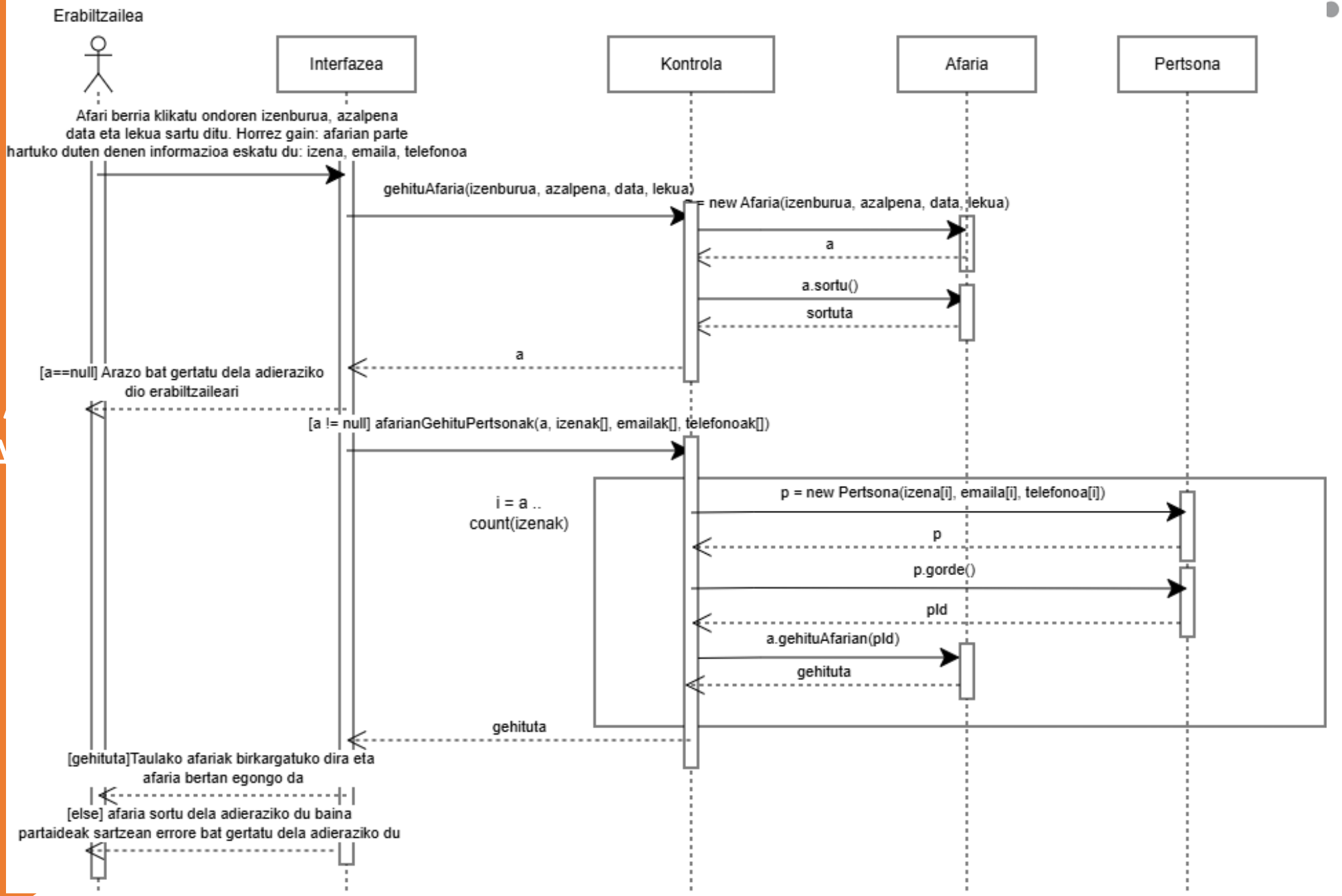
ADIBIDEA: AFARIA EZABATU





ADIBIDEA: AFARIAN IZENA EMAN

ADIBIDEA: A
IZENA EMA



```java
int result = JOptionPane.showConfirmDialog( parentComponent: this, panel,  title: "Izena eman afarian", JOptionPane.OK_CANCEL_OPTION, JOptionPa
if (result == JOptionPane.OK_OPTION) {
    String izena = izenaField.getText();
    String emaila = emailaField.getText();
    String telefonoa = telefonoaField.getText();

    if (!izena.isEmpty() && !emaila.isEmpty()) {
        try {
            int resultCode = controller.izenaEmanAfarian(afariaId, izena, emaila, telefonoa);
            switch (resultCode){
                case -1:
                case -3:
                    JOptionPane.showMessageDialog( parentComponent: this,  message: "Errorea pertsona sortzean",  title: "Errorea", JOptionPane.E
                    return;
                case -2:
                    JOptionPane.showMessageDialog( parentComponent: this,  message: "Pertsona dagoeneko afarian dago",  title: "Errorea", JOption
                    return;
                default:
                    JOptionPane.showMessageDialog( parentComponent: this,  message: "Izena eman duzu afarian!");
                    break;
            }
```

```java
public int izenaEmanAfarian(int afariaId, String izena, String emaila, String telefonoa) throws SQLException {
    //Pertsonaren id-a lortu
    int pertsonaId = Pertsona.bilatuPertsona(izena, emaila, telefonoa);

    Pertsona p = null;

    //Ez bada existitzen sortu pertsona
    if(pertsonaId <= 0) {
        //Pertsona ez da existitzen, sortu
        p = new Pertsona(izena, emaila, telefonoa);
        pertsonaId = p.sortu();
        if (pertsonaId < 0) {
            return -1; //Errorea gertatu da pertsona sortzean
        }
    }
```

```
        //Afarian badagoen begiratu
        if(Afaria.badagoPertsonaAfarian(afariaId, p)) {
            return -2; //Pertsona dagoeneko afarian
        }else {
            //Gehitu afarian
            if(Afaria.gehituPertsonaAfarian(afariaId, pertsonaId)) {
                return pertsonaId; //Ondo gehitu da
            } else {
                return -3; //Errorea gertatu da afarian gehitzean
            }
        }
    }
```

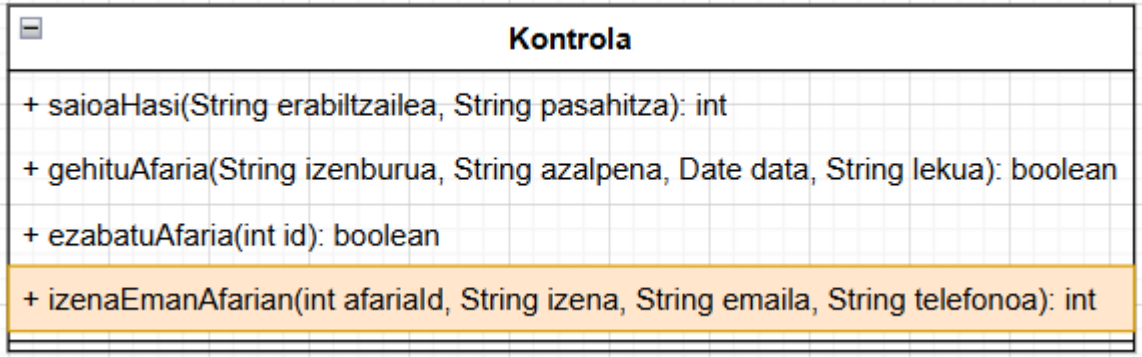
ADIBIDE
IZENA E

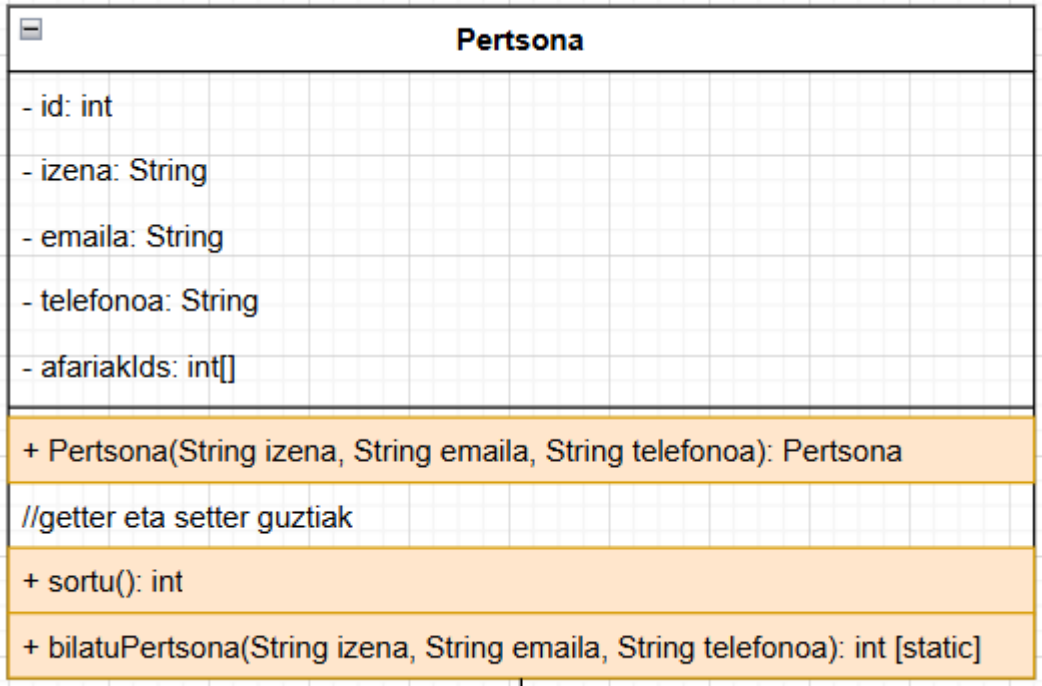
```java
public static int bilatuPertsona(String izena, String emaila, String telefonoa) throws SQLException {  1 usa
    String query = "SELECT id FROM partehartzaileak WHERE izena = ? AND emaila = ? AND telefonoa = ?";

    try (Connection conn = DbKonexioa.konektatu();
         PreparedStatement pstmt = conn.prepareStatement(query)) {

        pstmt.setString( parameterIndex: 1, izena);
        pstmt.setString( parameterIndex: 2, emaila);
        pstmt.setString( parameterIndex: 3, telefonoa);

        try (ResultSet rs = pstmt.executeQuery()) {
            if (rs.next()) {
                int id = rs.getInt( columnLabel: "id");
                return id;
            } else {
                return 0; // Ez da aurkitu
            }
        }
    } catch (SQLException e) {
        e.printStackTrace();
        return -1; // Errorea gertatu da
    }
```

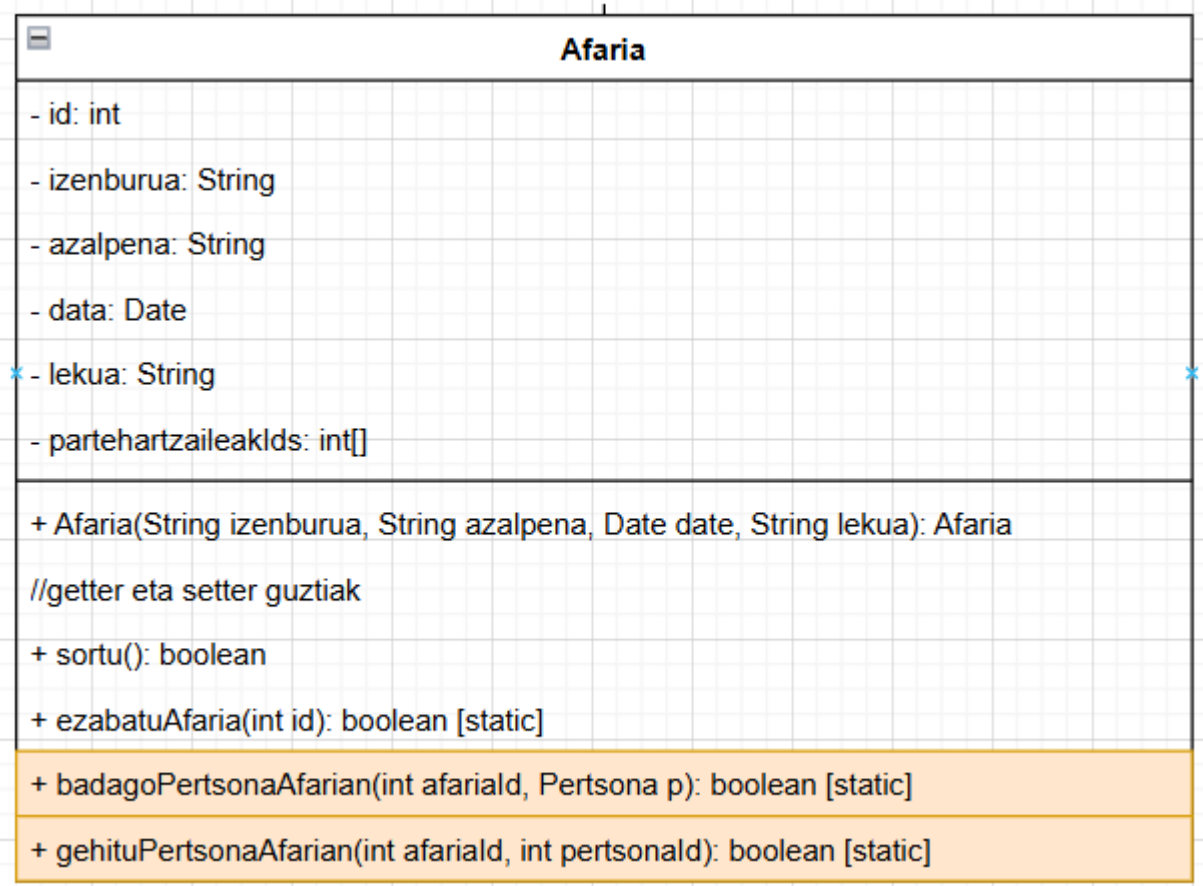
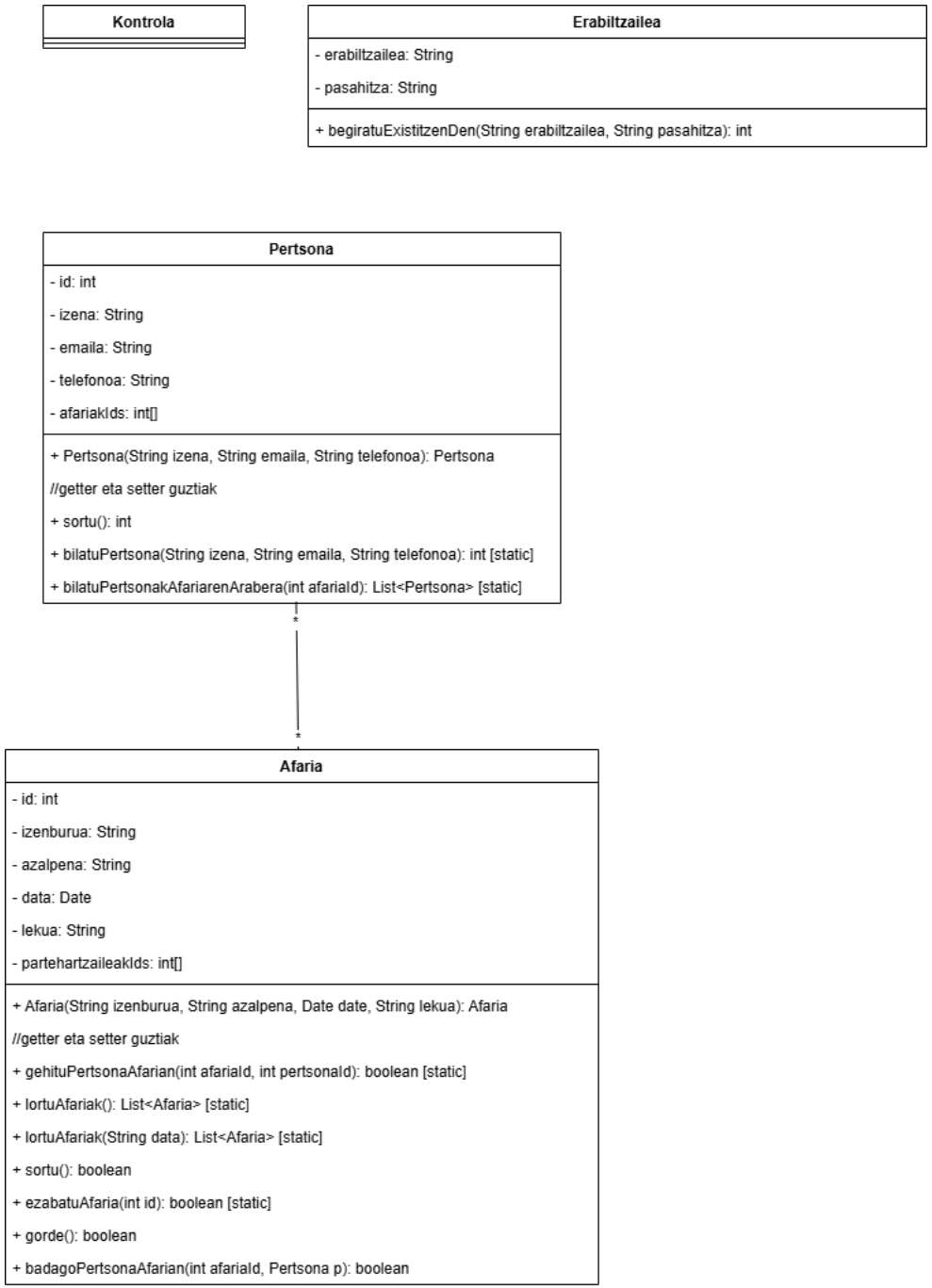
ADIBIDE
IZENA E

```java
public static boolean badagoPertsonaAfarian(int afariaId, Pertsona p) {  1 usage
    int pertsonaId = p.getId();
    String query = "SELECT * FROM afariak_partehartzaileak WHERE afaria_id = ? AND partehartzailea_id = ?";
    try (Connection conn = DbKonexioa.konektatu(); PreparedStatement pstmt = conn.prepareStatement(query)) {
        pstmt.setInt( parameterIndex: 1, afariaId);
        pstmt.setInt( parameterIndex: 2, pertsonaId);
        try (ResultSet rs = pstmt.executeQuery()) {
            return rs.next(); // Egiaztatzen du pertsona dagoen ala ez afarian
        }
    } catch (SQLException e) {
        e.printStackTrace();
        return false; // Errorea gertatu da
    }
}
```

Goierri eskola







ADIBIDEA: AMAIERAKO DIAGRAMA